**Lab 1 – AI$^2$ Software Requirements Specification**

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# 1. Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a detailed and structured description of the requirements for the Artificially Intelligent Invoices (AI²) system. This document serves as a technical reference for developers, outlining system functionality, interfaces, constraints, and assumptions necessary for implementation and validation.

## 1.2 Scope

AI² is a web-based system designed to automate portions of the Accounts Payable (AP) invoice processing workflow. The system retrieves invoices from designated email inboxes, extracts relevant data fields, and compares invoice data against corresponding purchase orders (POs) stored in an enterprise resource planning (ERP) system.

The system supports automated matching for invoices that meet predefined criteria and flags discrepancies, such as missing or inconsistent data, for human review. AI² maintains audit logs of all processing actions to support traceability and compliance requirements.

The system does not replace human decision-making in exception handling and does not perform financial transactions such as payment execution. It operates as a support tool within existing financial workflows and integrates with external systems but does not replace them.

## 1.3 Definitions, Acronyms and Abbreviations

**AI² (Artificially Intelligent Invoices):** The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.

**Alteryx:** A data automation and analytics tool that enables secure API connections and data workflows between applications.

**AP (Accounts Payable):** The amount a business owes for goods and services purchased on credit; typically, due at intervals of 30, 45, 60, or 90 days depending on repayment terms.

**APEX:** A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.

**Atlantic Diving Supply (ADS):** An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.

**AWS GovCloud:** A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.

**CMMC (Cybersecurity Maturity Model Certification):** A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.

**ERP (Enterprise Resource Planning System):** An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

**FedRAMP (Federal Risk and Authorization Management Program):** A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.

**Graph API:** A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.

**Invoice:** A document listing goods or services provided, including a statement of the amount due; a bill.

**Object-relational mapping (ORM)**: a programming technique that acts as a software layer translating data between relational databases and object-oriented programming languages.

**Oracle:** The system of record used by ADS for enterprise data and financial management.

**PO (Purchase Order):** The official order list ADS sends to suppliers detailing the items or services requested.

**Salesforce:** A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.

**Single Sign-On (SSO):** an authentication method that allows users to securely access multiple, connected applications and websites using a single set of credentials (username and password)

**Supplier:** A person or organization that provides goods or services needed by another organization.

**Vendor:** A person or company offering goods or services for sale.

## 1.4 References

ADS Pitch Document Sample

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### 1.5 Overview

Section 2, Overall Description, provides a high-level overview of the system, including its general functionality, operating environment, and constraints.

Section 3, Requirements Specification, defines the detailed functional and non-functional requirements of the system. This section is intended primarily for developers and serves as the basis for system design and implementation.

## 2. Overall Description

### 2.1 Product Perspective

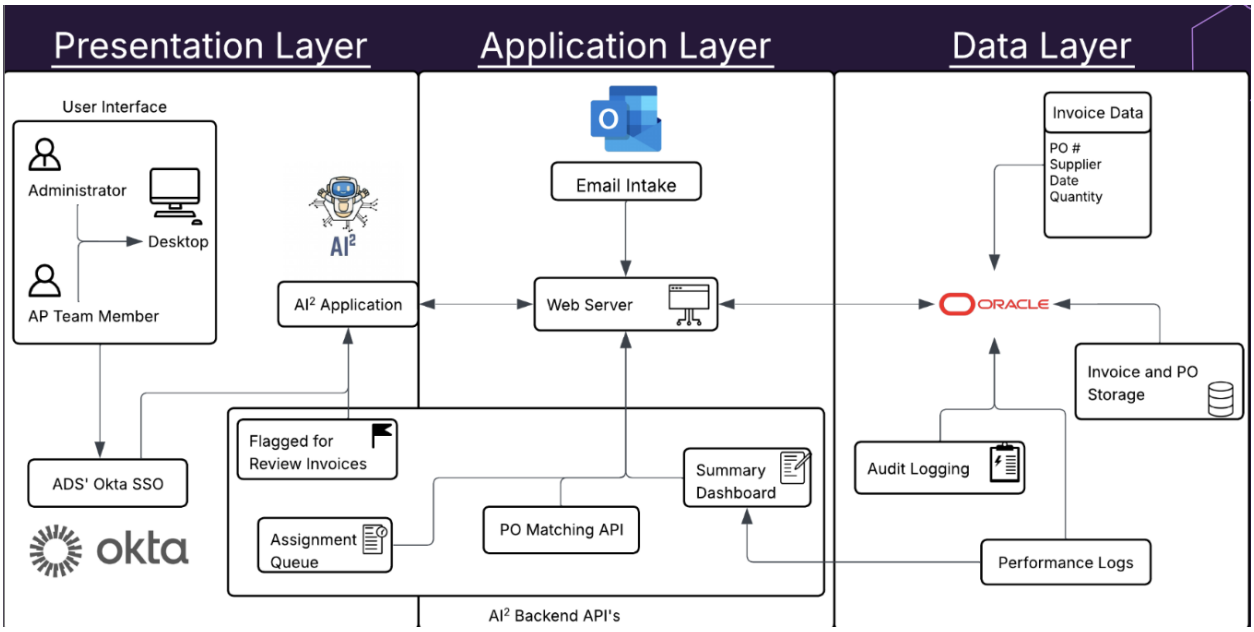
AI² is a web-based system accessed through modern desktop browsers by AP personnel and system administrators. The system operates within a secure cloud environment hosted on AWS GovCloud to meet federal security and compliance requirements. User authentication is managed through a Single Sign-On (SSO) solution, such as Okta, which is external to the system.

AI² functions as a subsystem within a larger enterprise environment. It integrates with external systems including email services for invoice retrieval and an ERP system for PO data. These external systems are not controlled by AI² but are essential for its operation.

The major architectural components of AI² are organized into three primary layers, as shown below in Figure 1.

**Figure 1**

*Major Functional Components Diagram*




The Presentation Layer provides the user interface through which AP personnel and administrators interact with the system. It supports role-based access, allowing users to review flagged invoices, monitor system activity, and manage workflows.

The Application Layer contains the core processing logic. It is responsible for retrieving invoices, extracting relevant data, performing invoice-to-PO matching, and routing exceptions for human review. This layer coordinates communication between the user interface, external services, and internal data storage.

The Data Layer manages persistent storage of invoice data, purchase order records, and system logs. It supports auditability by maintaining records of system decisions, user actions, and processing outcomes.

AI² follows a client-server architecture in which user requests are processed by backend services and results are returned to the user interface. The system relies on external services for

authentication, email retrieval, and enterprise data access, while maintaining internal control over processing logic and data management.

## 2.2 Product Functions

The AI² system provides a set of functional capabilities that support automated invoice processing, validation, and workflow management. These features are derived from the system requirements and are implemented to varying degrees within the current prototype. ***Table 1*** summarizes the primary system features and indicates their implementation status.

**Table 1**

*AI² Feature Description and Prototype Implementation*

| Features | Description | Prototype |
|---|---|---|
| Outlook Email Integration (Graph API) | Integrates with an email service to automatically retrieve incoming invoices from designated inboxes. | Eliminated |
| Invoice Parser and Data Extractor | Extracts structured data fields (e.g., purchase order number, invoice date, vendor, line items) from invoice documents. | Fully Implemented |
| Automated PO matching algorithm | Compare extracted invoice data with purchase order records to determine matches based on predefined criteria. | Partially Implemented (rule-based matching only) |

| | | |
|---|---|---|
| Invoice Quality Analyzer | Evaluates completeness and validity of extracted invoice data using a weighted scoring system. | Fully Implemented |
| Queue Assigning and Routing | Assigns invoices to processing queues based on supplier, priority, or business rules. | Partially Implemented (single queue only) |
| Queue Assignment Dashboard | Displays assigned invoices and their statuses to Accounts Payable personnel. | Partially Implemented (single dashboard view) |
| Model Training and Deployment Pipeline | Supports training and deployment of machine learning models for invoice processing. | Fully Implemented |
| Model Version Control | Maintains version history and tracking of trained models for reproducibility and rollback. | Fully Implemented |
| AI Performance Dashboard | Displays performance metrics of the AI system, including accuracy and processing efficiency. | Fully Implemented |
| Invoice Data Anonymization Pipeline | Removes or masks sensitive information from invoice data to support secure processing. | Eliminated (not required due to use of pre-anonymized data) |

| | | |
|---|---|---|
| Admin Configuration Dashboard | Allow administrators to configure system parameters such as confidence thresholds and routing rules. | Partially Implemented (limited configuration support) |
| System Health and Performance Dashboard | Monitors system performance metrics such as processing time and system load. | Partially Implemented (performance metrics only) |
| Error Logging and Recovery Module | Captures system errors and supports recovery mechanisms to maintain system stability. | Fully Implemented |
| Admin System Alert Notifications | Send notifications to administrators regarding system events or issues. | Partially Implemented (push notifications only) |
| Automated Supplier Notifications | Sends automated updates to suppliers regarding invoice status. | Eliminated |
| Automated AP Team Reminders | Notifies Accounts Payable personnel of pending tasks or deadlines. | Partially Implemented (push notifications only) |
| AI Confidence Scoring Module | Generates confidence scores indicating the reliability of automated invoice matching results. | Fully Implemented |

| | | |
|---|---|---|
| Continuous Learning from Human Interaction | Updates system behavior based on corrections made during human review. | Fully Implemented |
| Human Review Panel | Provides an interface for users to review, validate, and correct flagged invoices. | Fully Implemented |
| Encryption Verification Tool | Verifies that data is encrypted during transmission and storage. | Eliminated |
| AWS GovCloud Integration Tool | Supports deployment and operation within AWS GovCloud environments. | Eliminated |
| Audit Log Management System | Maintains logs of system actions and user interactions for traceability. | Partially Implemented |
| Compliance Changelog System | Tracks changes related to compliance requirements and system updates. | Partially Implemented |
| Auto Generated Compliance Report | Generates reports to demonstrate compliance with regulatory standards. | Eliminated |
| Compliance Regulations Monitoring Tool | Monitors changes in relevant compliance regulations. | Eliminated |
| Okta SSO Authentication Module | Integrates with Okta for secure user authentication and access control. | Eliminated |
| Alteryx-Oracle Integration Module | Facilitates data exchange between Alteryx workflows and Oracle systems. | Eliminated |

| | | |
|---|---|---|
| Integration Monitoring Dashboard | Displays status and health of external system integrations. | Eliminated |

The prototype focuses on core invoice processing and validation functionality, while several integration and compliance-related features are not included or only partially implemented.

## 2.3 User Characteristics

The AI² system is designed to support multiple user roles with varying levels of technical expertise and responsibilities within the Accounts Payable workflow. Administrators are responsible for system configuration, monitoring, and maintenance. They interact with system dashboards that provide visibility into performance, audit logs, and configuration settings. This role requires a moderate to high level of technical expertise, as administrators must manage system parameters and ensure proper operation. Their interaction with the system is periodic and typically occurs when configuration changes or system oversight is required.

Model trainers are responsible for developing, evaluating, and maintaining the machine learning models used within the system. This role requires a high level of technical expertise, including knowledge of machine learning concepts, data processing, and model evaluation techniques. Model trainers interact with the system infrequently, primarily during model updates, retraining processes, or performance improvements.

AP team members are the primary users of the system and interact with it daily. Their responsibilities include reviewing flagged invoices, validating extracted data, and resolving discrepancies identified by the system. This role requires a low to moderate level of technical expertise, as users primarily interact with the system through a structured user interface designed for operational efficiency.

Compliance officers are responsible for ensuring that the system adheres to regulatory and organizational standards. They review audit logs, verify compliance-related records, and assess system behavior against applicable requirements. This role requires a moderate level of technical understanding and is typically performed on a periodic basis, such as during audits or compliance reviews.

## 2.4 Constraints

The AI² system operates under several technical and regulatory constraints that influence its design and implementation. The system must function within a secure cloud environment that meets federal compliance standards, including requirements related to data protection and system security. Additionally, the system's ability to integrate with external services, such as email platforms and enterprise resource planning systems, is dependent on the availability, reliability, and compatibility of those systems' interfaces. The system must also support secure authentication mechanisms, including SSO, to ensure controlled access. Furthermore, all data handling and storage must comply with applicable organizational and regulatory security requirements. The current prototype is subject to additional constraints, as it does not include full integration with production-level enterprise systems and therefore operates within a limited, simulated environment.

## 2.5 Assumptions and Dependencies

The AI² system is developed based on several assumptions regarding its operating environment and available data. It is assumed that invoice data will be accessible through designated email inboxes and that purchase order data will be available from an external enterprise resource planning system. It is also assumed that users will have access to secure, internet-

connected workstations and modern web browsers capable of supporting the system interface. Additionally, the system assumes that incoming invoice data follows reasonably consistent formats that can be processed by the extraction and matching components.

The system also depends on several external services and technologies for full functionality. These dependencies include email services for invoice retrieval, enterprise resource planning systems for access to purchase order data, and a SSO provider for authentication and access control. The system further relies on cloud infrastructure to support hosting, storage, and secure operation. The availability and proper functioning of these external components are essential to the overall performance and reliability of the AI$^2$ system.